Efficient LLM Moderation with Multi-Layer Latent Prototypes

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The LLM Alignment Challenge

Post-Training Limits

While modern LLMs use RLHF and DPO for value alignment, robust moderation remains essential for deployment safety.

Superficial Safety

Research indicates safety alignment can be fragile, often losing effectiveness after only a few generated tokens.

Evolving Threats

Adversarial attacks and novel jailbreak techniques emerge constantly, bypassing static defenses.

Input vs. Output Moderation

Input Moderation (Filtering)

Proactive Defense

- Assesses the prompt before generation begins.
- Refusing malicious prompts early saves significant compute resources.
- Does not interfere with the model's internal reasoning process.

Output Moderation (Steering)

Reactive Control

- Analyzes or alters the response during or after generation.
- Often hinders model capabilities and reasoning on benign tasks.
- Consumes compute generating tokens that might be discarded.

Current Moderation Tools

Guard Models

Specialized LLMs evaluating inputs (e.g., LlamaGuard).

✓ **High Performance**
Very accurate at detecting harm.

✗ **High Cost**
Heavy compute and memory overhead.

✗ **Rigid**
Hard to customize without expensive fine-tuning.

Latent-Based Methods

Analyzing internal hidden states of the model.

✓ **High Efficiency**
Extremely fast, low memory.

✓ **Flexible**
Cheap to train on custom data.

✗ **Lower Performance**
Worse than Guard models.

Multi-Layer Prototypes (MLPM)

The MLPM solves the efficiency-performance dilemma by combining the best of both worlds.

Approach	Training Cost	Inference Efficiency	Flexibility	Safety Performance
Guard Models	High	Low	Low	High
Prior Latent Methods	Low	High	High	Medium
MLPM (Ours)	Low	High	High	High

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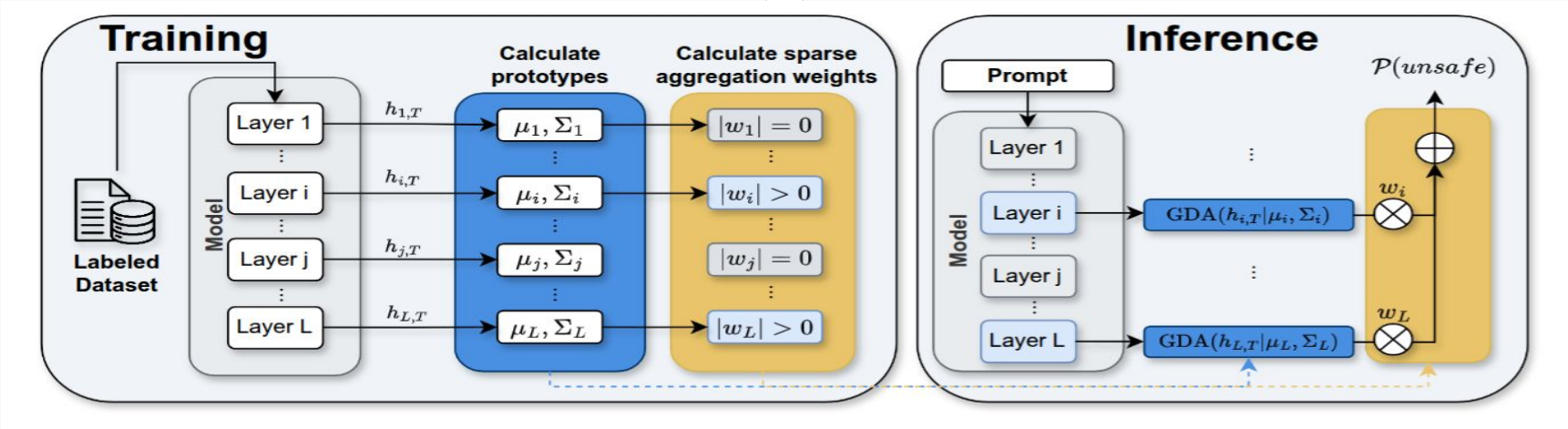
MLPM Architecture Overview

Phase 1: Training (Offline)

Extract hidden states *h* from each layer to calculate class prototypes (mean *μ* and covariance *Σ*).

Phase 2: Inference (Online)

For a new prompt, compute Mahalanobis distance at each layer and aggregate with sparse weights *w*.



Prototype-Based Classification (NMC)

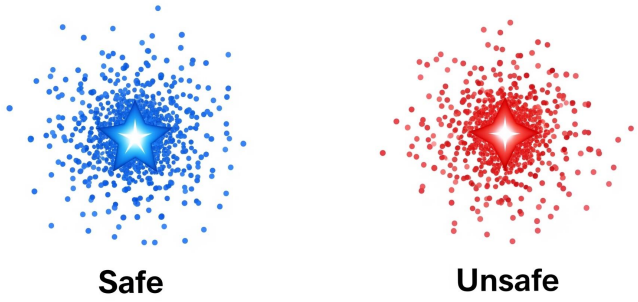
Methodology & Rationale

Why use Nearest Mean Classifier?

We utilize **NMC** due to its exceptional **generalization capabilities** and high **data-efficiency**.

Defining Prototypes

Prototypes are established as the **empirical means** of latent representations extracted for both **safe and unsafe** classes.



Mahalanobis Distance

Why not Euclidean?

Euclidean distance fails to capture the **complex geometry** of the latent space, assuming uniform variance in all directions.

The Mahalanobis Advantage

We utilize the **Mahalanobis distance** as it explicitly accounts for **data variance** and correlations within the distribution.

$$d_M(h, \mu_c, \Sigma_c) = \sqrt{(h - \mu_c)^\top \Sigma_c^{-1} (h - \mu_c)}$$

Estimating Safety Probabilities

Gaussian Discriminant Analysis (GDA)

We estimate class probabilities using GDA to map latent representations to a safety score.

Memory Optimization

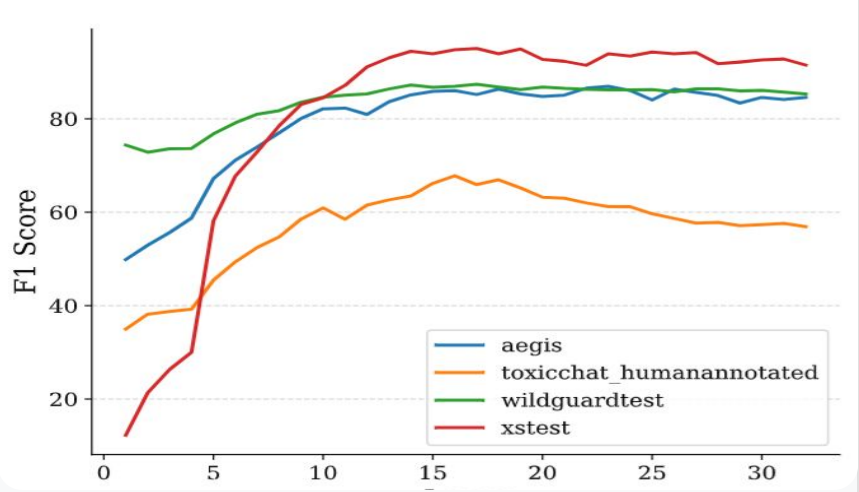
By utilizing a **shared covariance matrix** across all classes, we significantly reduce the computational footprint and memory usage during inference.

$$\mathcal{P}(c|x, \mu_c, \Sigma_c) = \frac{\exp(d_M(h, \mu_c, \Sigma_c))}{\sum_{i=1}^{k} \exp\left(d_M(h, \mu_i, \Sigma_i)\right)}$$

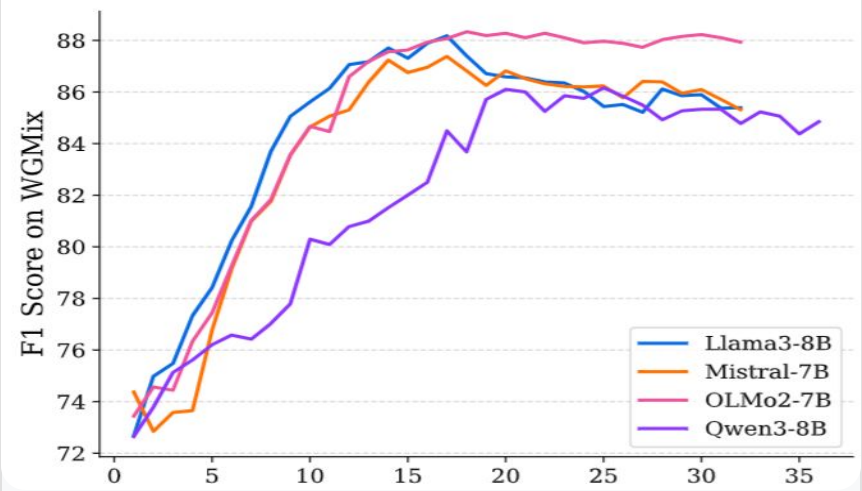
The Multi-Layer Advantage

Safety information is distributed across **intermediate layers**. Ambiguous inputs at a single layer are resolved by looking **holistically** across the model architecture.

F1 SCORE BY LAYER (Different Datasets)



F1 SCORE BY LAYER (Different Models)



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Sparse Aggregation Strategy

Optimizing compute by identifying and weighting the most informative transformer blocks.

METHODOLOGY

Feature Extraction

We extract final token representations from the feed-forward networks (FFNs) across *L* transformer blocks.

L1 Regularization

Aggregation uses learned weights with a penalty that encourages the model to rely on fewer, high-impact layers.

$$\mathrm{MLPM}(x) = \sigma\Big(\sum_{l=1}^{L} w^l \cdot \mathcal{P}^l(x \in \mathcal{X}_{\mathrm{harm}} | x, \mu^l, \Sigma^l)\Big)$$

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Experimental Setup

Comprehensive evaluation across diverse architectures and benchmark suites

Model Families Evaluated

We evaluate across diverse state-of-the-art model architectures to ensure cross-family robustness:

- **Llama**
- **Mistral**
- **OLMo**
- **Qwen3**

Benchmark Datasets

Harmfulness & Jailbreak (8 Datasets)

- Aegis, HarmBench, OpenAI Mod
- Simple Safety Tests, Toxic Chat
- XSTest, WildGuardMix, WildJailbreak

Neutral Benchmarks (Generalization)

- Alpaca, BigBenchHard, Codex
- GSM8k, MMLU, MTBench
- TruthfulQA

State-of-the-Art Performance

MLPM sets a new benchmark in efficiency and detection accuracy

Performance Benchmark Results

Dataset	Aegis	HarmB	OpenAI	SimpST	TChat	WGMix	WJB	XSTest	Average
Latent-Based									
Llama-8B-Inst+Ayub & Majumdar (2024)	82.52	96.98	66.60	98.48	55.62	80.91	82.85	92.76	82.09
Llama-8B-Inst+Abdelnabi et al. (2025)	84.16	95.18	67.99	98.99	59.59	86.27	93.27	90.63	84.51
Llama-8B-Inst+**MLPM(Ours)**	85.13	99.58	72.85	99.50	69.17	88.04	94.69	**97.44**	88.30
Qwen3-8B-Inst+Ayub & Majumdar (2024)	80.00	90.62	74.56	95.29	68.90	80.64	81.31	90.21	82.69
Qwen3-8B-Inst+Abdelnabi et al. (2025)	84.47	99.37	68.45	97.96	60.94	85.18	90.44	88.06	84.36
Qwen3-8B-Inst+**MLPM(Ours)**	83.49	**100.0**	72.35	96.91	64.40	86.21	92.00	92.23	85.95
Mistral-7B-Inst+Ayub & Majumdar (2024)	79.60	90.87	75.69	98.99	63.44	83.31	87.53	95.04	84.31
Mistral-7B-Inst+Abdelnabi et al. (2025)	84.55	97.86	64.59	98.48	57.63	85.73	91.13	94.60	84.32
Mistral-7B-Inst+**MLPM(Ours)**	87.36	99.16	70.68	99.50	66.33	87.63	93.65	96.10	87.55
OLMo2-7B-Inst+Ayub & Majumdar (2024)	88.11	96.54	66.95	**100.0**	65.63	88.09	96.84	94.33	87.06
OLMo2-7B-Inst+Abdelnabi et al. (2025)	84.16	93.30	75.19	99.50	72.53	86.20	93.48	94.43	87.35
OLMo2-7B-Inst+**MLPM(Ours)**	89.23	98.51	74.21	**100.0**	**76.51**	**88.52**	**97.55**	96.91	**90.18**
Guard Models									
Aegis-Guard-D (Ghosh et al., 2024)	81.00	70.46	76.44	97.96	75.61	72.09	75.44	81.53	78.82
LlamaGuard3 (Inan et al., 2023)	71.74	98.94	**79.11**	99.50	54.11	76.76	67.83	88.52	79.56
GraniteGuardian-3-1-8B (Padhi et al., 2024)	87.78	79.90	77.63	99.50	73.25	84.57	96.75	85.59	85.62
ShieldGemma-9B (Zeng et al., 2024)	77.44	69.04	77.63	91.30	68.13	58.88	59.94	82.41	73.10
WildGuard (Han et al., 2024)	**89.78**	99.37	72.28	99.50	70.14	88.04	97.10	95.26	88.93

Key Takeaways:

- MLPM **outperforms** prior latent-based methods across all tested architectures.
- MLPM **surpasses heavy guard models** such as LlamaGuard 3, ShieldGemma, and GraniteGuardian.
- Maintains **high performance** even when applied to lightweight model variants.

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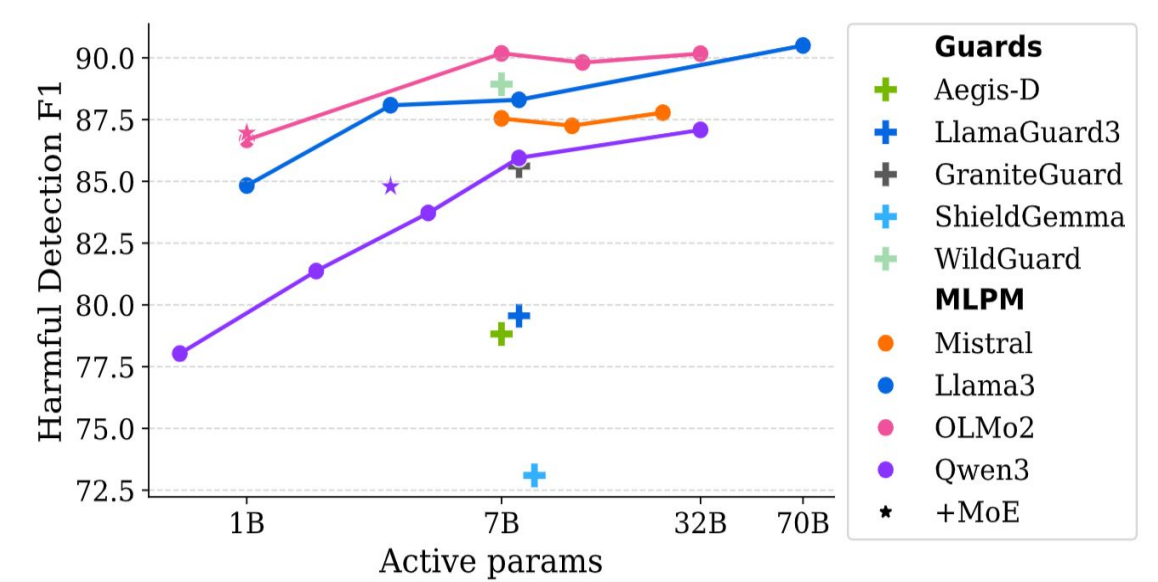
Scaling Across Model Sizes

MLPM's performance scales seamlessly with the model size.

HARMFUL DETECTION F1 VS. PARAMETER COUNT



Key Scaling Insights:

- Unlike fixed-size guard models, MLPM **performance increases** with the size of the backbone.
- Even Our **1B Llama3** variant (~85 F1) dramatically outperforms much larger fixed guards.

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MLPM Efficiency

MLPM introduces negligible overhead, making real-time safety enforcement practical

TRAINING EFFICIENCY

- Requires only a single forward pass without gradient calculations
- Prototype computation is fast enough to run on a standard CPU
- Remarkably data-efficient; effective even on small datasets

INFERENCE & MEMORY OVERHEAD

< 0.001%
Computational Overhead

~24 KB
Memory per Prototype

Negligible Latency: Overhead is minimal as intermediate representations are computed during prompt prefill.

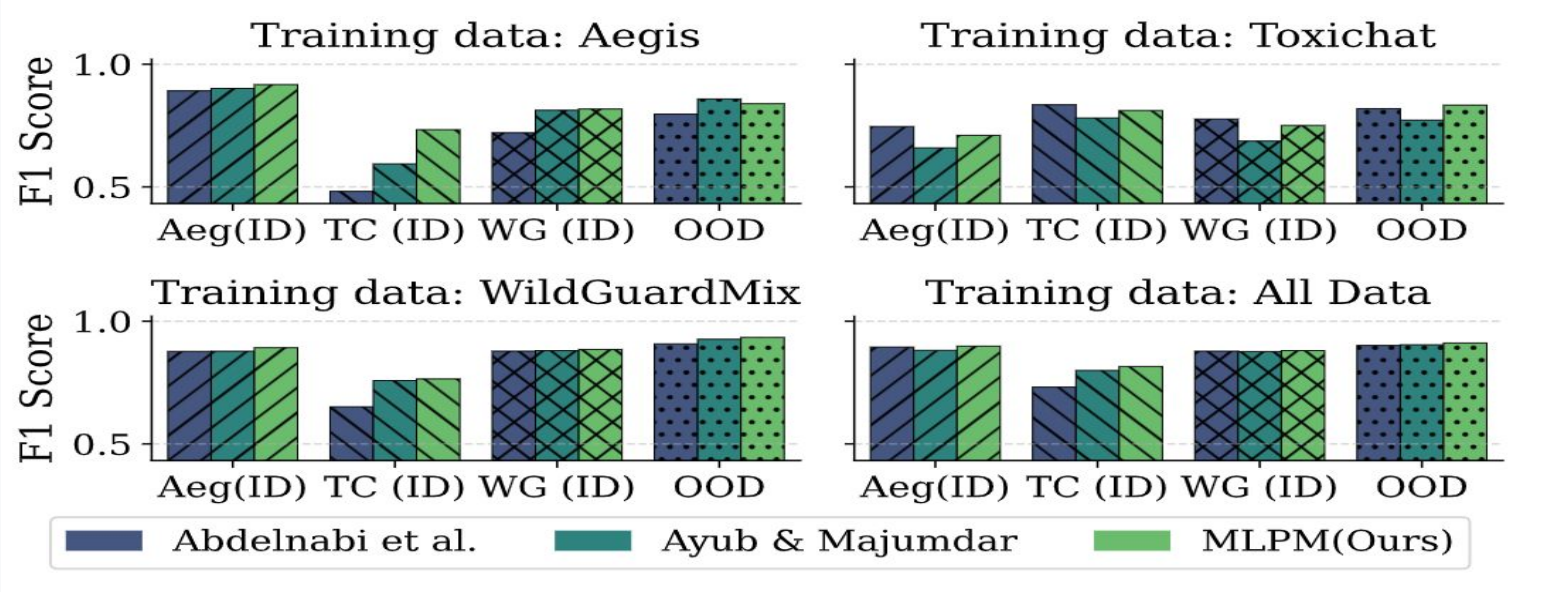
Highly Memory Efficient: Just 24KB per prototype (half-precision) for Llama 3.1-8B.

Practical Deployment: Ideal for real-time safety enforcement in production LLM environments.

Generalization (ID vs. OOD)

MLPM maintains high detection accuracy on out-of-distribution (unseen) threats.

PERFORMANCE ACROSS DATASETS



Comparison of F1 Scores across Aegis, Toxichat, WildGuardMix, and Combined Training Data

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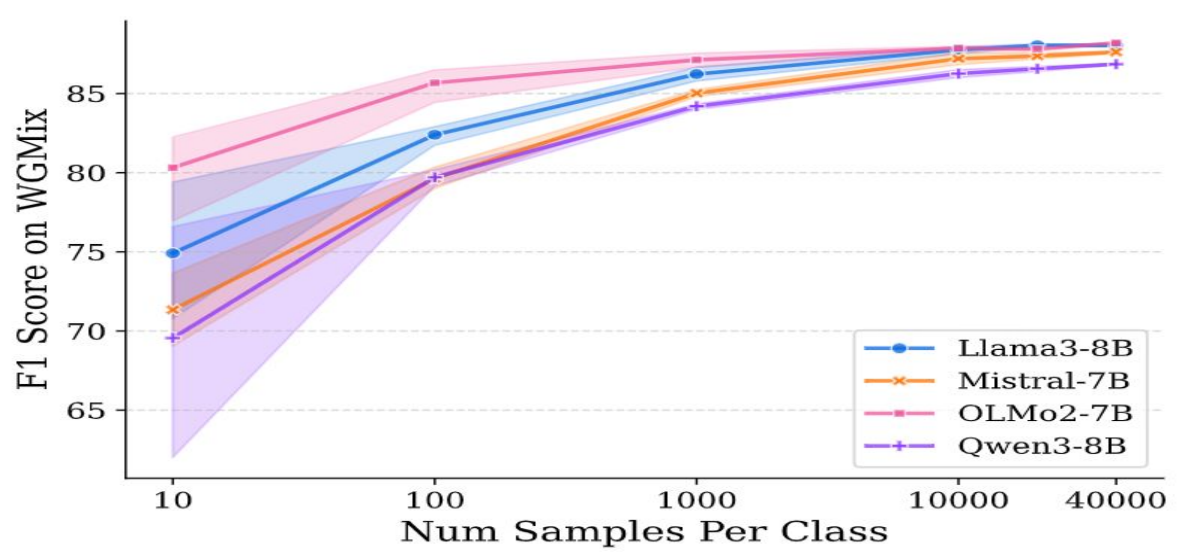
Exceptional Data Efficiency

MLPM achieves guard-level performance with as few as 1,000 samples, eliminating the need for massive datasets.

F1 SCORE VS. SAMPLES PER CLASS (WGMIX)



Performance trajectory of Llama3, Mistral, OLMo2, and Qwen3 across varying data scales

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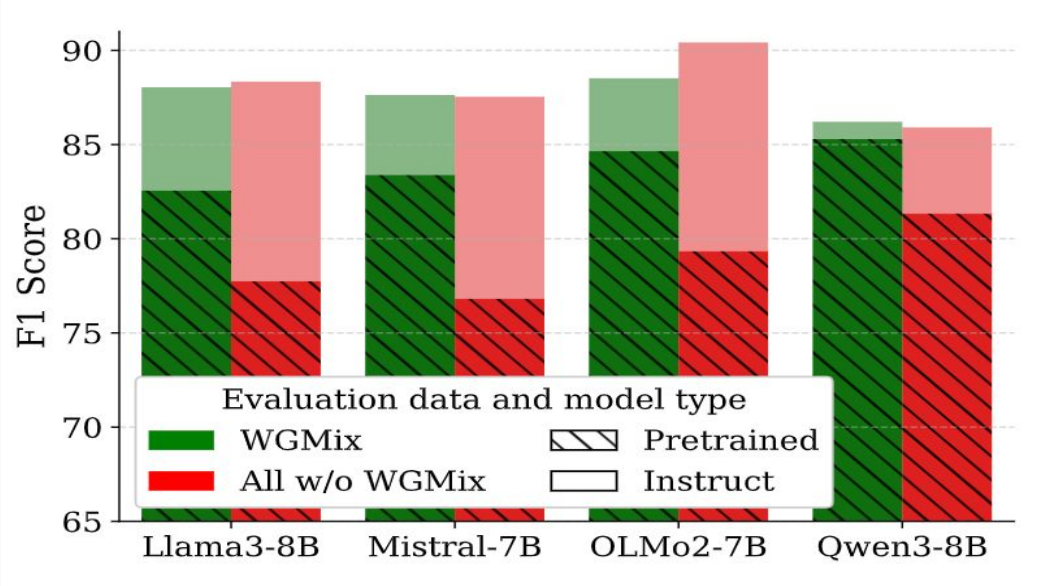
Pre-trained vs. Instruct Models

Instruction tuning is highly beneficial for MLPM, as pre-trained models struggle to generalize to OOD threats.

MODEL COMPARISON: F1 SCORE PERFORMANCE

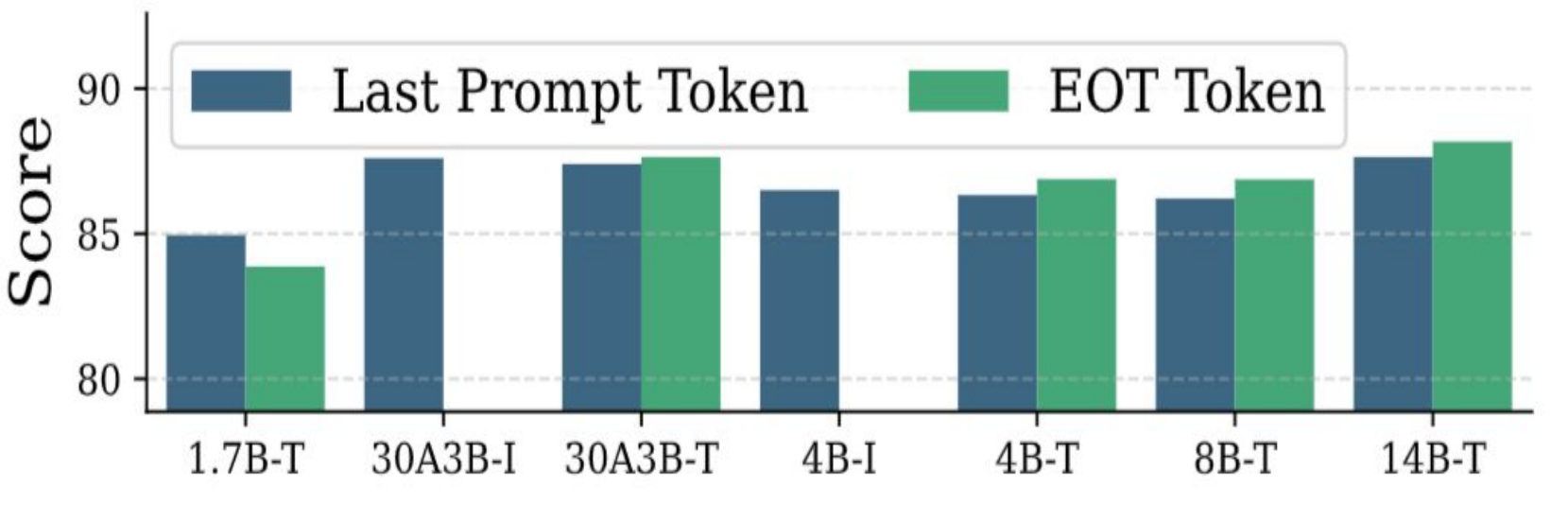


Comparative analysis of F1 Scores across Llama3, Mistral, OLMo2, and Qwen3 architectures

Impact of Chain-of-Thought on Detection

Comparing MLPM performance on the last prompt token vs. the final end-of-thinking (EOT) token across Qwen3 reasoning models.

Performance Score: Last Prompt Token vs. EOT Token



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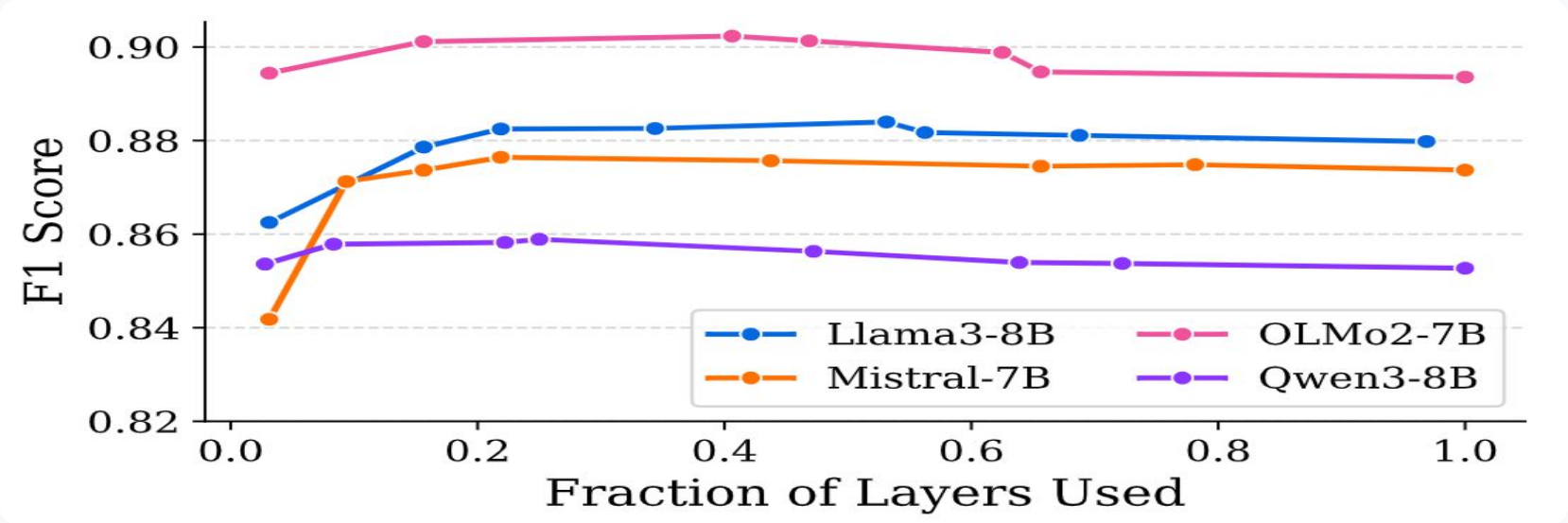
Optimal Performance via Sparse Subset Selection

Peak model performance is not achieved by using single layers or the full stack, but through a specific subset of layers.

F1 Score vs. Fraction of Layers Used across LLM Architectures



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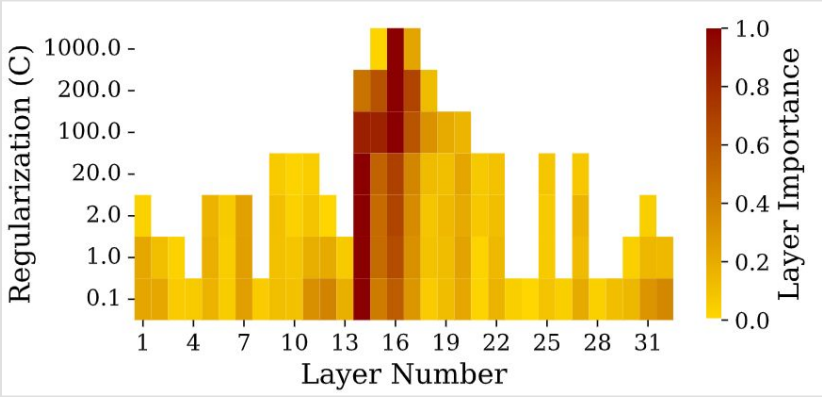
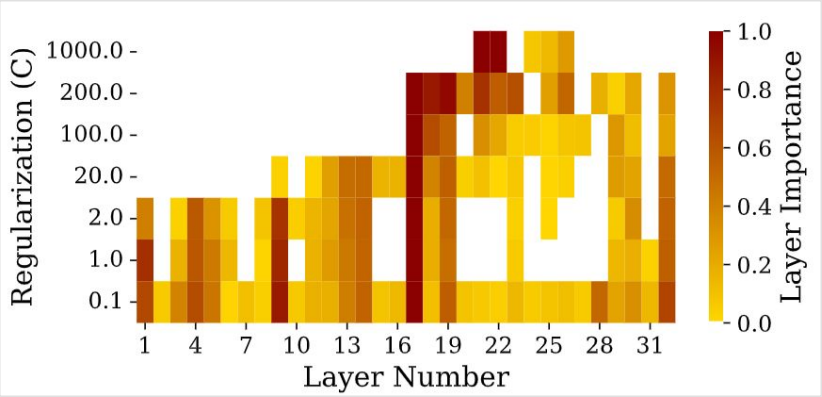
MLPM Layer Selection by Architecture

Safety representations vary wildly: Mistral (middle layers) vs. OLMo2 (final layers). MLPM automatically optimizes its subset selection.

Comparison of Layer Selection Distributions: Mistral-7B vs. OLMo2-7B



Mistral-7B: Middle Layers Concentration



OLMo2-7B: Late Layers Concentration

MLPM as a Conditioning Switch

Using MLPM to gate steering **dramatically reduces False Refusal Rates (FRR)** on safe prompts while maintaining robust moderation.

Performance for MLPM Conditioning across LLM Architectures

	Llama3-8B			Mistral-7B			OLMo2-7B			Qwen3-8B		
Method	**ASR↓**	**FRR↓**	**F1↑**	**ASR↓**	**FRR↓**	**F1↑**	**ASR↓**	**FRR↓**	**F1↑**	**ASR↓**	**FRR↓**	**F1↑**
Base Model	36.21	2.54	74.75	78.25	1.90	34.51	20.82	5.93	80.48	50.53	2.22	63.78
+MLPM Simple Refuse	15.52	5.93	**83.47**	15.92	6.24	**82.84**	14.59	6.03	**83.85**	18.30	6.24	**81.52**
Lee et al. (2025)	36.47	7.51	68.71	73.61	6.24	37.31	20.42	6.56	79.90	50.13	2.65	63.66
Turner et al. (2023)	35.94	53.76	27.20	40.45	18.73	54.29	18.57	8.15	78.90	47.75	3.81	64.41
+MLPM Conditioning	44.56	4.66	66.03	45.89	3.60	66.15	19.36	6.46	80.63	48.67	2.54	65.02
Zheng et al. (2024)	25.07	6.03	77.82	54.11	3.49	59.30	19.76	8.25	78.12	28.91	5.71	75.83
+MLPM Conditioning	26.92	4.02	79.21	55.44	2.43	59.16	20.95	6.35	79.86	31.56	3.92	76.30

- *Input moderation* and *output steering* act as complementary safety mechanisms.

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GDA Superiority in Robustness

Despite its simplicity, **GDA consistently outperforms** other single-layer methods across LLM architectures.

Robustness Comparison: GDA vs. Baseline Methods (F1 %)

Method	Llama-8B-Inst	Mistral-7B-Inst	OLMo2-7B-Inst	Qwen3-8B-Inst
GDA	**86.25**	84.18	**89.44**	**85.36**
Logistic Regression	84.51	**84.32**	86.82	84.36
MLP	85.44	83.93	87.24	84.02
Random Forest	84.31	82.09	87.63	82.69
XGBoost	86.02	82.15	89.37	84.47

Ablations: Distance and Layer Impact

Switching to **Mahalanobis distance** provides the largest performance gain, while **multi-layer aggregation** also provides significant boost.

Performance Ablation: Distance Metrics vs. Layer Representation

Distance	Prototypes	L-8B	M-7B	O-7B	Q-8B
Euclidean	Last layer	77.17	77.39	83.61	74.97
Euclidean	Multi-layer	83.73	80.38	85.74	78.96
Mahalanobis	Last layer	86.25	84.18	89.44	85.36
Mahalanobis	Multi-layer	**88.30**	**87.55**	**90.18**	**85.95**

Last Token vs. Mean Representation

Using the **last token** is significantly more effective for detecting prompt harmfulness, likely because it consistently captures the final contextual state.

Performance Comparison

Representation Used	Llama-8B	Mistral-7B	OLMo2-7B	Qwen3-8B
Last token	**88.30**	**87.55**	**90.18**	**85.95**
Mean representation	86.59	84.22	88.32	83.46

Error Analysis: Where Does MLPM Miss?

MLPM struggles mostly with **Social Stereotypes** and subtle **Sexual Content**, cases that are difficult even for humans.

Harm Category	% Misclassified by all models	Distribution
Social Stereotypes and Unfair Discrimination	23.68	
Others	20.41	
Sexual Content	19.15	
Fraud Assisting Illegal Activities	16.67	
Private Information Individual	14.81	
Copyright Violations	12.90	
Sensitive Information Organization Government	12.00	
Mental Health Over-Reliance Bias	10.64	
Benign	3.39	
Causing Material Harm By Disseminating Misinformation	2.22	
Cyberattack / False Misleading Info (Disinformation)	0.00	
Defamation / Toxic Hate Speech	0.00	
Violence and Physical Harm	0.00	

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Limitations of MLPM

While highly effective, MLPM is optimized as a specific component within a broader **safety architecture**.

LLM Dependency

MLPM's effectiveness is strictly bound by the quality and representation space of the underlying post-trained LLM.

System Role

Designed as a fast, first line of defense in a larger moderation system, not a standalone oracle for all edge cases.

Summary

State-of-the-Art Performance

Exceeds heavy guard models while maintaining a lightweight footprint for real-time applications.

Efficiency & Data Utilization

Negligible inference overhead and highly data-efficient, requiring minimal resources for high accuracy.

Seamless Scalability

Scales effortlessly with model size and adapts fluidly to any underlying neural architecture.

Questions?

Scan for Code

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